

Ollama LAN Access Guide

Access Ollama running on a Windows PC from a laptop on the same home network, including a Jio main router and Tenda AC1200 secondary router.

Known device	Network information
PC	192.168.0.196 / 255.255.255.0 / gateway 192.168.0.1
Laptop	192.168.29.21 / 255.255.255.0 / gateway 192.168.29.1
Ollama	Port 11434; embedding endpoint /api/embed
Example model	bge-m3

Goal: make the laptop send embedding requests to the PC at <http://192.168.0.196:11434/api/embed>.

The guide covers the full path from Ollama configuration through network diagnosis. The final network fix depends on whether the laptop is on the same subnet as the PC or is separated by the Tenda router.

1. Understand localhost vs LAN access

When Ollama is accessed as `http://localhost:11434`, localhost means the same computer that is making the request. A laptop cannot use its own localhost to reach the PC. The laptop must connect to the PC using the PC's LAN IP address.

For this setup the target PC address is:

```
http://192.168.0.196:11434/api/embed
```

Ollama must also listen on a network interface, not only loopback.

2. Set OLLAMA_HOST on Windows

On the PC running Ollama, open PowerShell and set the user environment variable permanently:

```
[Environment]::SetEnvironmentVariable("OLLAMA_HOST", "0.0.0.0:11434", "User")
```

Verify the saved value:

```
[Environment]::GetEnvironmentVariable("OLLAMA_HOST", "User")
```

Expected output:

```
0.0.0.0:11434
```

Temporary alternative: for one PowerShell session, use `$env:OLLAMA_HOST="0.0.0.0:11434"` before starting ollama serve.

3. Restart Ollama correctly

Fully stop the existing Ollama process before testing the new binding.

```
taskkill /IM ollama.exe /F
```

Then start Ollama again. If the desktop app is not being used, you can start the server manually:

```
ollama serve
```

Keep that terminal open while testing when using manual server mode.

4. Verify Ollama is listening on the LAN

On the PC, run:

```
netstat -ano | findstr :11434
```

Correct output should contain a listener such as:

```
TCP    0.0.0.0:11434    0.0.0.0:0        LISTENING
```

An IPv6 listener such as [::]:11434 is also fine.

A result showing only 127.0.0.1:11434 or [::1]:11434 means Ollama is bound only to loopback and is not yet reachable from another machine.

In this case the confirmed working output was:

```
TCP    0.0.0.0:11434    0.0.0.0:0        LISTENING
TCP    [::]:11434      [::]:0           LISTENING
```

This means Ollama itself is listening correctly for LAN traffic.

5. Test Ollama locally on the PC

```
curl http://localhost:11434/api/tags
```

Then test the PC LAN address from the PC itself:

```
curl http://192.168.0.196:11434/api/tags
```

Both should return the installed Ollama model list. The second command confirms that the LAN-bound listener is usable through the PC's actual address.

6. Windows Firewall test

If the laptop cannot connect, allow TCP port 11434 through Windows Firewall. Open PowerShell as Administrator:

```
New-NetFirewallRule `
  -DisplayName "Ollama 11434" `
  -Direction Inbound `
  -Protocol TCP `
  -LocalPort 11434 `
  -Action Allow
```

To check the rule:

```
Get-NetFirewallRule -DisplayName "Ollama 11434"
```

For a private LAN, you can also scope the rule to the Private profile:

```
New-NetFirewallRule `
  -DisplayName "Ollama 11434" `
  -Direction Inbound `
  -Protocol TCP `
  -LocalPort 11434 `
  -Action Allow `
  -Profile Private
```

Turning off the firewall temporarily is useful only as a diagnostic. If connectivity still fails with the firewall completely disabled, the problem is not the Windows firewall.

7. Test connectivity from the laptop

From the laptop, test the PC port directly:

```
Test-NetConnection 192.168.0.196 -Port 11434
```

The key field is:

```
TcpTestSucceeded : True
```

If True, test Ollama itself:

```
curl http://192.168.0.196:11434/api/tags
```

Then test the embedding endpoint:

```
curl http://192.168.0.196:11434/api/embed `
  -Method POST `
  -ContentType "application/json" `
  -Body '{"model":"bge-m3","input":["Hello world"]}'
```

8. Diagnose the actual two-router network

The device addresses reveal the important part of the topology:

Device	IPv4	Gateway	DHCP server
PC	192.168.0.196	192.168.0.1	192.168.0.1
Laptop	192.168.29.21	192.168.29.1	192.168.29.1

This means the PC and laptop are on different IPv4 subnets. The PC is on the Jio/main router subnet, while the laptop is receiving its address from the Tenda subnet.

The physical arrangement you described is:

```
Jio Home Router
|
+---- Wi-Fi ---- Laptop (192.168.29.21)
|
+---- Ethernet ---- Tenda AC1200 ---- Ethernet ---- PC (192.168.0.196)
```

The IP/DHCP evidence is more important than the physical cable description: the PC is currently receiving DHCP from 192.168.0.1, so the PC is logically on the Jio subnet. The laptop is logically behind 192.168.29.1.

9. Test whether the laptop can reach the two networks

From the laptop:

```
ping 192.168.29.1
ping 192.168.0.196
```

If the first succeeds and the second times out, the Tenda is reachable but traffic from the laptop subnet to the Jio/PC subnet is being blocked or not routed.

You can also inspect the path:

```
tracert 192.168.0.196
```

10. Recommended solution: use the same LAN

For a simple home setup, the cleanest solution is to have the laptop and PC on the same LAN. There are two practical approaches.

Approach A - connect the laptop to the same Jio LAN

If the laptop can connect directly to the Jio Wi-Fi rather than the Tenda Wi-Fi, do that. After reconnecting, run:

```
ipconfig
```

The laptop should then receive something like 192.168.0.xxx with gateway 192.168.0.1. Test:

```
ping 192.168.0.196
curl http://192.168.0.196:11434/api/tags
```

If these work, the Ollama service is ready for the laptop.

Approach B - use the Tenda as an access point

If the Tenda is only needed to provide Ethernet/Wi-Fi in the room, AP/Access Point/Bridge mode is usually the simplest architecture. Both devices then live on the Jio network:

```
Jio Router 192.168.0.1
|
+---- Laptop 192.168.0.x
|
+---- Tenda in AP mode
      |
      +---- PC 192.168.0.x
```

In this design the laptop can directly use:

```
http://192.168.0.196:11434/api/embed
```

This avoids inter-subnet routing and NAT entirely.

11. Keeping the Tenda as a separate router

If you intentionally want the laptop to remain on 192.168.29.0/24 and the PC to remain on 192.168.0.0/24, the setup requires routing and/or port forwarding. The laptop cannot simply reach the PC as if both were on the same LAN.

A typical routed design would look like:

```
Laptop 192.168.29.21
      |
      v
Tenda LAN 192.168.29.1
      |
Tenda WAN 192.168.0.x
      |
      v
Jio Router 192.168.0.1
      |
      v
PC 192.168.0.196
```

Depending on the Tenda firmware and exact WAN/LAN configuration, you may need:

Setting	Typical requirement
Tenda port forwarding	TCP 11434 -> 192.168.0.196:11434 if the request enters through the Tenda WAN side
Routing	A route between 192.168.29.0/24 and 192.168.0.0/24, depending on which direction traffic must
Firewall	Allow traffic on the relevant interface(s) and TCP 11434

Because router firmware differs, do not blindly apply a port-forward rule until the Tenda's exact model and WAN/LAN IPs are known.

12. Python configuration for the laptop

Your RAG application should replace localhost with the PC address:

```
import requests

OLLAMA_URL = "http://192.168.0.196:11434/api/embed"

response = requests.post(
    OLLAMA_URL,
    json={
        "model": "bge-m3",
        "input": text_list
    }
)

response.raise_for_status()
embeddings = response.json()["embeddings"]
```

The laptop is now the client, while the PC performs the embedding inference through its Ollama instance and GPU/CPU.

13. Final verification checklist

Check	Expected result
OLLAMA_HOST	0.0.0.0:11434
netstat	0.0.0.0:11434 LISTENING
PC local API	curl localhost:11434/api/tags works
PC LAN API	curl 192.168.0.196:11434/api/tags works
Laptop port test	TcpTestSucceeded : True
Laptop API test	curl 192.168.0.196:11434/api/tags works
Embedding test	/api/embed returns embeddings
Python	Use http://192.168.0.196:11434/api/embed

Troubleshooting order

Do not change multiple things at once. Work from the server outward:

1. OLLAMA_HOST -> 0.0.0.0:11434
2. Restart Ollama
3. netstat -> 0.0.0.0:11434 LISTENING
4. Test PC LAN address locally
5. Test laptop port 11434
6. Test laptop /api/tags
7. Test laptop /api/embed
8. If port test fails, investigate firewall/routing/router isolation
9. If possible, put both devices on the same 192.168.0.x LAN

Current confirmed addresses from the setup

PC: 192.168.0.196
Gateway: 192.168.0.1

Laptop: 192.168.29.21
Gateway: 192.168.29.1

Ollama: 0.0.0.0:11434

Key point: Ollama was successfully configured to listen on the LAN. The remaining obstacle is the network path between the laptop subnet (192.168.29.0/24) and the PC subnet (192.168.0.0/24).

End of guide.